

2.47

- (a) $P(B) = P(A \cap B) + P(A^c \cap B) = P(B|A)P(A) + P(B|A^c)P(A^c) = 0.2 \cdot 0.7 + 0.4 \cdot (1 - 0.7) = 0.14 + 0.12 = 0.260$
- (b) If A and B are independent, then $P(A \cap B) = P(A)P(B) = 0.4 \cdot 0.3 = 0.12 \neq 0$. Thus they cannot be mutually exclusive.

2.50

Let A be the event that the transferred ball is yellow and B the event that the ball from Box 2 is yellow.

(a)

$$\begin{aligned} P(B) &= P(A \cap B) + P(A^c \cap B) = P(B|A)P(A) + P(B|A^c)P(A^c) \\ &= \frac{2}{3} \cdot \frac{2}{6} + \frac{1}{3} \cdot \frac{4}{6} = \frac{4}{9} \end{aligned}$$

(b)

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)} = \frac{(2/3)(2/6)}{4/9} = \frac{1}{2}$$

2.54

Let A be the event that she is late for the physics course and B the event that she is late for the statistics course.

- (a) $P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B) - P(A)P(B) = 0.4 + 0.3 - (0.4)(0.3) = 0.58$
- (b) $P[(A \cup B)^c] = 1 - P(A \cup B) = 1 - 0.58 = 0.42$
- (c) $P(A \cup B) - P(A \cap B) = P(A \cup B) - P(A)P(B) = 0.58 - 0.3 \cdot 0.4 = 0.46$

2.57

Let B be the event that George replaces the battery, T the event that he replaces the starter, and R the event that the car runs.

- (a) $P(R) = P(R|B)P(B) + P(R|T)P(T) = (0.8)(0.9) + (0.15)(0.1) = 0.735$
- (b)

$$\begin{aligned} P(T|R^c) &= \frac{P(T \cap R^c)}{P(R^c)} = \frac{P(R^c|T)P(T)}{1 - P(R)} = \frac{[1 - P(R|T)]P(T)}{1 - P(R)} = \frac{[1 - 0.15](0.1)}{1 - 0.735} \\ &= \frac{0.085}{0.265} = 0.321 \end{aligned}$$

2.58

Let R be the event that a car is recalled, and S_i the event that a car is from Site i , $i = 1, 2, 3$. Then

$$P(R) = \sum_{i=1}^3 P(R|S_i)P(S_i) = (0.04)(0.5) + (0.06)(0.3) + (0.08)(0.2) = 0.054$$

(a)

$$P(S_1|R) = \frac{P(R|S_1)P(S_1)}{P(R)} = \frac{(0.04)(0.5)}{0.054} = 0.37$$

(b)

$$P(S_2|R) = \frac{P(R|S_2)P(S_2)}{P(R)} = \frac{(0.06)(0.3)}{0.054} = 0.33$$

(c)

$$P(S_3|R) = \frac{P(R|S_3)P(S_3)}{P(R)} = \frac{(0.08)(0.2)}{0.054} = 0.30$$

2.60

Let R be the event that a refrigerator lasts more than 15 years, and A, B, C the events that a refrigerator is from company A, B, C, respectively. Then

$$(a) \quad P(R) = P(R|A)P(A) + P(R|B)P(B) + P(R|C)P(C) = (0.1)(0.5) + (0.2)(0.3) + (0.05)(0.2) = 0.12$$

(b)

$$P(B|R) = \frac{P(R|B)P(B)}{P(R)} = \frac{(0.2)(0.3)}{0.12} = 0.5$$

3.3

(a)

x	Event	$P(X = x)$
2	$\{(1,1)\}$	$1/16$
3	$\{(1,2), (2,1)\}$	$1/8$
4	$\{(1,3), (2,2), (3,1)\}$	$3/16$
5	$\{(1,4), (2,3), (3,2), (4,1)\}$	$1/4$
6	$\{(2,4), (3,3), (4,2)\}$	$3/16$
7	$\{(3,4), (4,3)\}$	$1/8$
8	$\{(4,4)\}$	$1/16$

(b)

$$F(x) = \begin{cases} 0 & \text{if } x < 2 \\ 1/16 & \text{if } 2 \leq x < 3 \\ 3/16 & \text{if } 3 \leq x < 4 \\ 3/8 & \text{if } 4 \leq x < 5 \\ 5/8 & \text{if } 5 \leq x < 6 \\ 13/16 & \text{if } 6 \leq x < 7 \\ 15/16 & \text{if } 7 \leq x < 8 \\ 1 & \text{if } x \geq 8 \end{cases}$$

(c)

$$P(3 \leq X < 7) = P(3 \leq X \leq 6) = F(6) - F(2) = \frac{13}{16} - \frac{1}{16} = \frac{12}{16} = \frac{3}{4}$$

3.8

(a)

$$F(x) = \begin{cases} 0 & \text{if } x < 1 \\ 0.01 & \text{if } 1 \leq x < 2 \\ 0.03 & \text{if } 2 \leq x < 3 \\ 0.15 & \text{if } 3 \leq x < 4 \\ 0.4 & \text{if } 4 \leq x < 5 \\ 0.82 & \text{if } 5 \leq x < 6 \\ 0.98 & \text{if } 6 \leq x < 7 \\ 1 & \text{if } x \geq 7 \end{cases}$$

(b)

$$\begin{aligned} E(X) &= \sum_{x=1}^7 xf(x) \\ &= 0.01 + 2(0.02) + 3(0.12) + 4(0.25) + 5(0.42) + 6(0.16) + 7(0.02) = 4.61 \end{aligned}$$

(c)

$$\begin{aligned} E(X^2) &= \sum_{x=1}^7 x^2 f(x) \\ &= 0.01 + 4(0.02) + 9(0.12) + 16(0.25) + 25(0.42) + 36(0.16) + 49(0.02) = 22.41 \end{aligned}$$

$$Var(X) = E(X^2) - \mu^2 = 22.41 - 4.61^2 = 1.1579$$

(d) $P(X < 5) = 0.4 < 0.75 < P(X \leq 5) = 0.82$. Hence the third quartile is 5.

3.10

(a)

x	0	1	2	3	4
$f(x)$	0.05	0.25	0.3	0.3	0.1

(b)

$$E(X) = \sum_{x=0}^4 xf(x) = 0 + 0.25 + 2(0.3) + 3(0.3) + 4(0.1) = 2.15$$

$$E(X^2) = \sum_{x=0}^4 x^2 f(x) = 0 + 0.25 + 4(0.3) + 9(0.3) + 16(0.1) = 5.75$$

$$Var(X) = E(X^2) - \mu^2 = 5.75 - 2.15^2 = 1.1275$$

$$\sigma = \sqrt{1.1275} = 1.0618$$

(c) $P(1.5 < X \leq 3) = f(2) + f(3) = 0.3 + 0.3 = 0.6$

(d) $P(X < 2) = f(0) + f(1) = 0.05 + 0.25 = 0.3$

(e) $P(X = 2) = f(2) = 0.3$

(f) $P(X > 2) = f(3) + f(4) = 0.3 + 0.1 = 0.4$

(g) $F(2.5) = P(X \leq 2.5) = f(0) + f(1) + f(2) = 0.05 + 0.25 + 0.3 = 0.6$

(h) $P(X \geq 3) = f(3) + f(4) = 0.3 + 0.1 = 0.4$

(i) $P(X < 2) = 0.3 < 0.5 < P(X \leq 2) = 0.6$. Hence the median is 2.